

CAGR Proof Pack

This pack provides categorical proof that the published CAGR values are derived directly from the live total return series and reconcile exactly to an independent recomputation.

Formula

$$\text{CAGR} = ((\text{Ending Index Level} / \text{Starting Index Level}) ^ {1 / \text{Year Fraction}}) - 1$$

Year Fraction used in the live pipeline verification: exact elapsed days / 365

Verification Summary

Series	Status	Start Date	End Date	Day Count	Start Value	End Value	Published CAGR	Independent CAGR	Match
AM100	Live benchmark series	2016-11-01	2026-01-30	3377	1,000.000000	27,055.008768	42.82%	42.82%	Yes
AM200	Live benchmark series	2019-05-02	2026-01-30	2465	1,000.000000	18,380.672941	53.89%	53.89%	Yes
AM300	Live benchmark series	2018-05-01	2026-01-30	2831	1,000.000000	17,570.004485	44.71%	44.71%	Yes
EAC25	Research-only historical series	2017-02-01	2026-01-30	3285	1,000.000000	2,036.224057	8.22%	8.22%	Yes
EAC_EXT	Research-only historical series	2017-02-01	2026-01-30	3285	1,000.000000	3,554.404588	15.13%	15.13%	Yes

Methodology Note

The published CAGR is derived directly from each live series total return index file. This verification pack recomputes CAGR independently from the first and last index levels over the exact elapsed day count. The return methodology itself is unchanged; where published CAGR has moved after data corrections, the change reflects revised constituent ranking, selection, and weighting rather than a change in formula.

Interpretation note: AM100, AM200, and AM300 are live benchmark series. EAC25 and EAC_EXT are research-only historical series and should not be presented as institutional benchmark track records.

Chain of Calculation

AM100

Step	Detail
1. Load total return series	/Users/derrythornalley/AM100_DATA/output/AM100_total_return.csv
2. Read first and last valid observation	2016-11-01 -> 2026-01-30
3. Measure exact elapsed days	3377
4. Convert to year fraction using 365-day convention	9.252054794521
5. Apply CAGR formula	$((27055.008768291300/1000.000000000000)^{(1/9.252054794521)}) - 1 = 42.824660022554\%$
6. Compare to published metrics file	Published 42.824660022554%; delta 0.000000000000e+00

AM200

Step	Detail
1. Load total return series	/Users/derrythornalley/AM100_DATA/output/AM200_total_return.csv
2. Read first and last valid observation	2019-05-02 -> 2026-01-30
3. Measure exact elapsed days	2465
4. Convert to year fraction using 365-day convention	6.753424657534
5. Apply CAGR formula	$((18380.672941315352/1000.000000000000)^{(1/6.753424657534)}) - 1 = 53.892627576509\%$
6. Compare to published metrics file	Published 53.892627576509%; delta 0.000000000000e+00

AM300

Step	Detail
1. Load total return series	/Users/derrythornalley/AM100_DATA/output/AM300_total_return.csv
2. Read first and last valid observation	2018-05-01 -> 2026-01-30
3. Measure exact elapsed days	2831

Step	Detail
4. Convert to year fraction using 365-day convention	7.756164383562
5. Apply CAGR formula	$((17570.004484818863/1000.0000000000000)^{(1/7.756164383562)}) - 1 = 44.706511042085\%$
6. Compare to published metrics file	Published 44.706511042085%; delta 0.000000000000e+00

EAC25

Step	Detail
1. Load total return series	/Users/derrythornalley/AM100_DATA/output/EAC25_total_return.csv
2. Read first and last valid observation	2017-02-01 -> 2026-01-30
3. Measure exact elapsed days	3285
4. Convert to year fraction using 365-day convention	9.000000000000
5. Apply CAGR formula	$((2036.224056948332/1000.0000000000000)^{(1/9.0000000000000)}) - 1 = 8.221600275546\%$
6. Compare to published metrics file	Published 8.221600275546%; delta 0.000000000000e+00

EAC_EXT

Step	Detail
1. Load total return series	/Users/derrythornalley/AM100_DATA/output/EAC_EXT_total_return.csv
2. Read first and last valid observation	2017-02-01 -> 2026-01-30
3. Measure exact elapsed days	3285
4. Convert to year fraction using 365-day convention	9.000000000000
5. Apply CAGR formula	$((3554.404587994412/1000.0000000000000)^{(1/9.0000000000000)}) - 1 = 15.132071273267\%$
6. Compare to published metrics file	Published 15.132071273267%; delta -8.326672684689e-17